

# sloggs — MIDI Implementation Chart

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This document describes the MIDI behavior of the **sloggs** MIDI implementation, an emulation of the Microsoft GS Wavetable Synth.

Updates along with [SPEC.adoc](#).

## 1. Basic Specifications

### Number of Parts

16.

### Maximum Polyphony

54 voices (48 sustained under continuous overload)

### Effects

None.

### Wavetable source

A DLS-1 sample collection. (Depicted here is the standard **gm.dls**.)

### Output format

22050 Hz, stereo, 16-bit signed PCM.

## 2. Receive data

In every byte pattern below, **n** is the MIDI channel nibble (**0-F**, **0**=channel 1 ... **F**=channel 16).

### 2.1. Channel Voice Message

#### 8n kk vv — Note off

Releases the note on its patch's own release curve, at the same rate in decibels per second no matter how far the note had already decayed — so a note released late fades out sooner, not more slowly.

- **kk** = note number (**00-7F**, 0-127)
- **vv** = release velocity (**00-7F**) — **ignored**

#### 9n kk vv — Note on

- **kk** = note number (0-127)
- **vv** = velocity (**01-7F**, 1-127) - fixed curve. Perceptually, amplitude scales with the square of velocity/127

Velocity is not only a volume control: on patches whose sample collection asks for it, a harder-struck note also **attacks** faster. In **gm.dls** 27 of the 235 instruments do. Note number behaves the same way for decay — see [Envelopes](#).

#### An kk vv — Polyphonic Key Pressure (aftertouch)

**Not recognized.** Received, no audible effect.

#### Bn cc vv — Control Change

See [Control Change](#).

#### Cn pp — Program Change

- **pp** = program number (**00-7F**, 0-127)

Combines with whatever Bank Select value is currently in effect ([Control Change](#)) to select an instrument.

#### Dn vv — Channel Pressure (aftertouch)

**Not recognized.** Received, no audible effect.

#### En ll mm — Pitch Bend Change

- **ll** = LSB
- **mm** = MSB
- Combine to a 14-bit value  $mm \times 128 + ll$ .
- Center is **00 40** (8192, no bend)
- Applied continuously to every sounding note on the channel — a bend arriving mid-note is heard **immediately**, not just on the next note. Range is set by RPN0, default  $\pm 2$  semitones ([Control Change](#)).

### 2.2. Control Change

Only the controllers below are recognized. **Every other CC number is received and produces no effect, silently — never an error.**

#### Bn 00 mm — Bank Select MSB

See Bank Select LSB.

#### Bn 20 ll — Bank Select LSB

**GS Mode only:** Takes effect on the **next Program Change**, never on notes already sounding.

### WARNING

GS Mode CCs are **dropped outright** if GS mode is off — the value is not stored, not queued, not remembered.

GS mode is entered only by a GS Reset **System Exclusive**, and is left again by GM System On, GM System Off or System Reset (FF)

#### Bn 01 vv — Modulation Wheel

Recognized.

#### Bn 06 mm — Data Entry MSB

See Data Entry LSB.

#### Bn 26 11 — Data Entry LSB

Recognized only together with RPN 0/1/2 selection, below; meaningless on their own.

#### Bn 07 vv — Channel Volume

Recognized. Default 64 (100).

#### Bn 0A vv — Pan

Recognized. Default 40 (64, center).

#### Bn 0B vv — Expression

Recognized. Default 7F (127).

#### Bn 40 vv — Hold 1 (Sustain)

Recognized.

- 00 = off (pedal up) — default
- 01–7F = on (pedal down). **Any non-zero value**, not just 40.

#### Bn 5B vv — Effects 1 Depth (Reverb Send)

**Not recognized.** Received w/o effect.

#### Bn 5D vv — Effects 3 Depth (Chorus Send)

**Not recognized.** Received w/o effect.

#### Bn 62 11 — NRPN LSB

See NRPN MSB.

#### Bn 63 mm — NRPN MSB

Recognized only as "deselect the current RPN". **No NRPN parameter is ever addressable** — Data Entry sent after an NRPN selection has no effect.

Sending either, with any value, immediately clears whatever RPN was previously selected to Null.

#### Bn 64 11 — RPN LSB

See RPN MSB.

#### Bn 65 mm — RPN MSB

Selects a Registered Parameter for the following Data Entry (CC6/CC38) messages to target:

| RPN (MSB,LSB) | Parameter                                                                                                                                                                                                       | Applied                                                                                   |
|---------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------|
| 00 00         | <b>Pitch Bend Sensitivity.</b><br><br><b>Data MSB only:</b><br>Semitones (default=2).<br><br><b>Not range-checked</b> — any MSB 00–7F is accepted. Bends are still capped at ±48 semitones.<br><br>LSB ignored. | Live — read continuously, alongside Pitch Bend itself.                                    |
| 01 00         | <b>Channel Fine Tuning</b><br><br>14-bit Data Entry:<br>–100 ... +99 cents.                                                                                                                                     | <b>Latched at Note On.</b><br>Changing it while a note is held does not retune that note. |
| 02 00         | <b>Channel Coarse Tuning</b><br><br><b>Data MSB only:</b><br>–64 ... +63 semitones.<br><br>LSB ignored.                                                                                                         | <b>Latched at Note On,</b><br>same as Fine Tuning.                                        |
| 7F 7F         | <b>Null</b> (no parameter selected).                                                                                                                                                                            | —                                                                                         |
| Others        | <b>Any</b>                                                                                                                                                                                                      | Stored, but <b>no effect</b>                                                              |

#### Bn 78 00 — All Sound Off

Recognized. Releases (does not hard-cut) every sounding note on the channel, **bypassing** a held sustain pedal.

#### Bn 79 00 — Reset All Controllers

Recognized Bank/Program and the RPN/NRPN selection are left untouched, otherwise re-arms:

| Parameter      | Value         |
|----------------|---------------|
| Channel Volume | 100           |
| Pan            | 64 (center)   |
| Expression     | 127           |
| Pitch Bend     | 8192 (center) |
| Modulation     | 0             |
| Sustain pedal  | Released      |

### WARNING

Reset All Controllers (CC121), GS Reset, GM System On/Off and System Reset (FF) **DO NOT reset** Pitch Bend Sensitivity (RPN0), nor which RPN is currently selected — both survive every reset path.

Fine Tuning (RPN1) and Coarse Tuning (RPN2) are **not** in that exception: CC121 leaves them alone, but GS Reset, GM System On/Off and System Reset (FF) all zero them.

#### Bn 7A 00 — Local Control

**Not applicable.** Received, no effect.

#### Bn 7B 00 — All Notes Off

Recognized. Releases every sounding note on the channel, **honoring** a held sustain pedal (notes under a held pedal stay held).

**Bn 7C 00 — Omni Mode Off**

See Omni Mode On.

**Bn 7D 00 — Omni Mode On**

**Not recognized.** Received, no mode change ([Channel Mode Message](#)).

**Bn 7E mm — Mono Mode On (Poly Off)**

Recognized. Switches the channel to mono and **releases** every currently-sounding note on it (implicit all-notes-off, per the MIDI spec), bypassing a held sustain pedal exactly as All Sound Off does.

mm (requested mono voice count) is **ignored** — the value sent changes nothing.

**Bn 7F 00 — Poly Mode On**

Recognized. Switches the channel to poly and **releases** every currently-sounding note on it (implicit all-notes-off), bypassing a held sustain pedal.

**2.3. Channel Mode Message**

Mode is only ever **Omni Off** (each channel answers exactly the Part(s) mapped to it — see Basic Channel in the [chart](#), below). Omni On/Off messages are received but never change this.

Mono/Poly are switchable per-channel via CC126/CC127 ([Control Change](#)); a channel is otherwise Poly by default.

**2.4. System Common Message**

Song Position Pointer, Song Select and Tune Request: **not recognized**. s**l**o**p**g**s** renders a Standard MIDI File on its own clock and has no use for sequencer-sync messages.

**2.5. System Real Time Message**

**F8 Timing Clock, FA Start, FB Continue, FC Stop**  
**Not recognized.**

**FE Active Sensing**  
**Not recognized.** Received, no effect.

**FF Reset**  
Recognized. Restores the power-on defaults below. Note that it does **not** restore everything — see the warning after the table.

| Parameter                          | Value         |
|------------------------------------|---------------|
| Bank/Program                       | 0 / 0 / 0     |
| Channel Volume                     | 100           |
| Pan                                | 64 (center)   |
| Expression                         | 127           |
| Pitch Bend                         | 8192 (center) |
| Modulation                         | 0             |
| Sustain pedal                      | Released      |
| Mono/Poly                          | Poly          |
| Fine / Coarse Tuning (RPN1 / RPN2) | 0 / 0         |
| Data Entry                         | 0             |
| Master Volume                      | Unity         |
| Per-Part tuning grid               | Flat (0)      |

| Parameter          | Value                                                       |
|--------------------|-------------------------------------------------------------|
| Part → channel map | Back to default (see <a href="#">Implementation Chart</a> ) |
| Rhythm parts       | Part 0 only                                                 |
| GS mode            | Cleared (Bank Select inert until the next GS Reset)         |

**WARNING**

The currently-selected **RPN/NRPN parameter** and the **Pitch Bend Sensitivity (RPN0)** value survive a Reset unchanged.

**Fine/Coarse Tuning (RPN1/RPN2)** do not — a Reset zeroes both.

**2.6. System Exclusive Message**

Four SysEx envelopes are recognized. The first is Roland's **DT1** ("Data set 1"), which carries one parameter address per message and is expanded in [Roland Messages](#) below; the other three are the standard Universal messages.

|                                         |                                                                                                                                                                                                                                                        |
|-----------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| F0 41 dev 42 12 aa bb cc dd...dd sum F7 | Roland GS-family message. <ul style="list-style-type: none"><li>dev = device ID — <b>not checked</b></li><li>aa bb cc = 21-bit parameter address.</li><li>dd...dd = data bytes for that address.</li><li>sum = checksum — <b>not checked</b></li></ul> |
| F0 7E dev 09 01 F7                      | GM System On (Universal Non-Realtime).<br><br>Full device reset, and <b>clears GS mode</b> — Bank Select goes inert again until the next GS Reset. dev not checked.                                                                                    |
| F0 7E dev 09 02 F7                      | GM System Off (Universal Non-Realtime).<br><br>Full device reset, and <b>clears GS mode</b> — exactly as GM System On does. dev not checked.                                                                                                           |
| F0 7F dev 04 01 11 mm F7                | Master Volume (Universal Realtime).<br><br>dev not checked. <b>Only mm (MSB) is used; 11 (LSB) is never read.</b>                                                                                                                                      |

**2.6.1. Roland Messages**

The addresses listed below are DT1 addresses, to be filled in the aa bb cc part of the Roland GS-family message above.

| Address  | Parameter                 | Effect                                                                                                                                                                                        |
|----------|---------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 40 00 7F | GS Reset                  | Full device reset, <b>and</b> sets GS mode.                                                                                                                                                   |
| 40 1n 02 | RCV CHANNEL (GS Mode)     | Remaps which incoming channel Part n listens to. Persists until a Reset.                                                                                                                      |
| 40 1n 15 | USE RHYTHM PART (GS Mode) | Nonzero = Part n switches to drum-kit ("rhythm") mode.<br><br><b>Also fires a Reset All Controllers on every channel</b> — all notes cut, controllers back to defaults. Bank/Program survive. |

| Address         | Parameter                                                                                                                                       | Effect                                                                           |
|-----------------|-------------------------------------------------------------------------------------------------------------------------------------------------|----------------------------------------------------------------------------------|
| 40 1n 40–4B     | Scale Tuning<br><br>One signed offset per semitone class (C...B), centred on 40, stored per Part n                                              | Recognized. Twelve data bytes are consumed, always starting at semitone class C. |
| 40 1n any other | Anything else in Roland's Patch-parameter address space for Part n (Reverb/Chorus Macro, Part Level, Pan, Pitch Key Shift, Velocity Sense, ...) | <b>Not individually implemented.</b>                                             |

#### NOTE

Only 40 00 7F (GS Reset) works from a cold start. Everything else is **ignored unless GS Mode is activated via said Reset.**

#### Not recognized:

- System Parameters (Master Tune, Master Volume-via-DT1, Master Key-Shift, Master Pan — address family 40 00 00–06)
- Display Data (address family 10 00 xx, Model ID 45)
- Drum Setup Parameters (address family 41 xx xx — per-drum-instrument key/level/pan/reverb/chorus edits)
- Bulk Dump (address families 48 xx xx/49 xx xx)

Any DT1 (or RQ1) message with an address outside the table above, or with a manufacturer ID other than 41/7E/7F, is accepted and silently dropped.

Fallthrough 40 1n xx.. messages ("not individually implemented") are instead consumed by the same 12-byte write that serves Scale Tuning, regardless of which parameter the address nominally names, and always starting at semitone class C. **A sequence intended to set, e.g., Reverb Macro or Part Level on real GS hardware will overwrite that Part's whole Scale Tuning table instead.**

## 3. MIDI Implementation Chart

#### Basic Channel

The MIDI channel each of the 16 Parts answers to. Parts are numbered 0–15, matching the n in the 40 1n xx System Exclusive addresses.

#### Mode

Refers to the MIDI Mode messages (Omni On/Off, Mono/Poly). s1opgs always behaves as Omni Off; only the Mono/Poly half of Mode is switchable, per channel.

#### Note Number / True Voice

The range of note numbers accepted, and the range actually reproduced. Note number 60 is middle C.

#### Velocity

Whether Note On and Note Off messages' velocity byte affects anything.

#### After Touch

KeyIs = polyphonic (per-note) aftertouch, ChIs = channel (monophonic) aftertouch.

#### Control Change

See Control Change for the full per-controller list — this row just points there.

#### Program Change

The program numbers a Program Change message can select (raw message value — one less than the corresponding instrument number as usually printed on an instrument list).

#### System Exclusive

See System Exclusive Message for the full address table.

#### System Common, System Real Time

Sequencer/drum-machine sync messages. Not used, except Reset (FF).

### 3.1. Implementation Chart

| Func                  | Transmit | Accept                 | Remarks                                                                                                                                                                                                                                 |
|-----------------------|----------|------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Basic Channel Default | ×        | 1 – 16                 | Not an identity map. See the <a href="#">default parts mapping</a> below.                                                                                                                                                               |
| Basic Channel Changed | ×        | 1 – 16, each           | Via GS RCV CHANNEL — which is itself inert until a GS Reset.                                                                                                                                                                            |
| Mode Default          | ×        | Mode 3                 | Omni Off, Poly.                                                                                                                                                                                                                         |
| Mode Messages         | ×        | Mono, Poly (CC126/127) | Omni On/Off received, no effect.                                                                                                                                                                                                        |
| Mode Altered          | ×        | Mode 3, 4              |                                                                                                                                                                                                                                         |
| Note Number           | ×        | 0 – 127                | Depends on the loaded gm.d1s.                                                                                                                                                                                                           |
| True Voice            | ×        | 0 – 127                | Depends on the loaded gm.d1s.                                                                                                                                                                                                           |
| Pitch Bender          | ×        | 0                      | 14-bit, continuous.                                                                                                                                                                                                                     |
| Control Change        | ×        | see below              | Its own table, directly below this one.                                                                                                                                                                                                 |
| Prog Change           | ×        | 0                      |                                                                                                                                                                                                                                         |
| True #                | ×        | 0 – 127                |                                                                                                                                                                                                                                         |
| System Exclusive      | ×        | 0                      | <ul style="list-style-type: none"> <li>• GM On/Off</li> <li>• GS Reset</li> <li>• RCV CHANNEL</li> <li>• USE RHYTHM PART</li> <li>• Master Volume</li> <li>• Scale Tuning</li> <li>• <b>Device ID and checksum unchecked</b></li> </ul> |
| System Common         | ×        | ×                      |                                                                                                                                                                                                                                         |
| System Real Time      | ×        | ×                      | Except Reset — see Aux Messages, below.                                                                                                                                                                                                 |
| <b>Aux Messages</b>   |          |                        |                                                                                                                                                                                                                                         |
| Local ON/OFF          | ×        | ×                      |                                                                                                                                                                                                                                         |
| All Notes OFF         | ×        | 0 (CC123)              |                                                                                                                                                                                                                                         |
| Active Sense          | ×        | ×                      |                                                                                                                                                                                                                                         |
| Reset                 | ×        | 0 (FF)                 |                                                                                                                                                                                                                                         |
| <b>Velocity</b>       |          |                        |                                                                                                                                                                                                                                         |
| Note ON               | ×        | 0                      |                                                                                                                                                                                                                                         |
| Note OFF              | ×        | ×                      |                                                                                                                                                                                                                                         |

| Func        | Transmit | Accept | Remarks |
|-------------|----------|--------|---------|
| After Touch |          |        |         |
| Key's       | ×        | ×      |         |
| Ch's        | ×        | ×      |         |

3.2. Control Change

| CC                    | Transmit | Accept | Remarks                                                                 |
|-----------------------|----------|--------|-------------------------------------------------------------------------|
| Bank Select<br>0 / 32 | ×        | cond.  | No effect until a GS Reset has been received.                           |
| Mod Wheel<br>1        | ×        | O      |                                                                         |
| Data Entry<br>6 / 38  | ×        | O      | Only meaningful with RPN, below.                                        |
| Channel Volume<br>7   | ×        | O      | Default 100.                                                            |
| Pan<br>10             | ×        | O      | Default 64 (center).                                                    |
| Expression<br>11      | ×        | O      | Default 127.                                                            |
| Hold 1<br>64          | ×        | O      |                                                                         |
| Reverb Send<br>91     | ×        | ×      | No reverb processor exists.                                             |
| Chorus Send<br>93     | ×        | ×      | No chorus processor exists.                                             |
| NRPN<br>98 / 99       | ×        | cond.  | Only deselects the current RPN; no NRPN parameter is ever addressable.  |
| RPN<br>100 / 101      | ×        | O      | Pitch Bend Sensitivity / Fine Tune / Coarse Tune only.                  |
| All Sound Off<br>120  | ×        | O      | Bypasses a held sustain pedal.                                          |
| Reset All Ctrl<br>121 | ×        | O      | Bank/Program and RPN selection untouched.                               |
| Local Control<br>122  | ×        | ×      | No local keyboard to disconnect.                                        |
| All Notes Off<br>123  | ×        | O      | Honors a held sustain pedal.                                            |
| Omni Off<br>124       | ×        | ×      |                                                                         |
| Omni On<br>125        | ×        | ×      |                                                                         |
| Mono On<br>126        | ×        | O      | Also releases the channel (implicit all-notes-off, bypasses the pedal). |
| Poly On<br>127        | ×        | O      | Also releases the channel (implicit all-notes-off, bypasses the pedal). |
| Other<br>—            | ×        | ×      | No effect, no error.                                                    |

NOTE

- 0 : recognized/handled.
- × : received (or n/a) with no effect.
- cond. : recognized, but only under a stated condition —

see Remarks.

Mode 1: Omni On, Poly.  
Mode 2: Omni On, Mono.  
Mode 3: Omni Off, Poly.  
Mode 4: Omni Off, Mono.

3.3. Default Parts Mapping

On boot, the parts-channels mapping are set as follows:

| Part | Channel | Assignment |
|------|---------|------------|
| 0    | 10      | Rhythm     |
| 1    | 1       | Melodic    |
| 2    | 2       | Melodic    |
| 3    | 3       | Melodic    |
| 4    | 4       | Melodic    |
| 5    | 5       | Melodic    |
| 6    | 6       | Melodic    |
| 7    | 7       | Melodic    |
| 8    | 8       | Melodic    |
| 9    | 9       | Melodic    |
| 10   | 11      | Melodic    |
| 11   | 12      | Melodic    |
| 12   | 13      | Melodic    |
| 13   | 14      | Melodic    |
| 14   | 15      | Melodic    |
| 15   | 16      | Melodic    |

These mappings can be changed via the [Roland SysEx Messages RCV CHANNEL](#) and [USE RHYTHM PART](#).

4. Miscellaneous Quirks

4.1. Polyphony

An automatic accelerated-release policy engages from the **49th simultaneous voice** — not at 54 — freeing voices faster than a plain steal-oldest policy would.

A voice cut this way, or cut by a drum kit's exclusive-key group, reaches silence within 14 ms of the cut however long its patch's own release is; it fades, never clicks off.

4.2. Envelopes

Amplitude fades at a constant rate in decibels, so a note fades at the same slope whatever level it is released from, and a patch's authored decay and release times mean "the time to fall 96 dB". Both are also **scaled per note** by the sample collection: higher notes decay faster, and on some patches harder-struck notes attack faster.

5. Presets

NOTE

All tables in this section depicts Microsoft's **gm.dls**, **slopps** does not provide one. Your chosen DLS file will vary.

5.1. Instruments

| #     | Name (bank 0) | Variants (bank MSB: name) |
|-------|---------------|---------------------------|
| Piano |               |                           |

| #                           | Name (bank 0) | Variants (bank MSB: name)                             |
|-----------------------------|---------------|-------------------------------------------------------|
| 0                           | Piano 1       | 8: Piano 1<br>16: Piano 1d                            |
| 1                           | Piano 2       | 8: Piano 2                                            |
| 2                           | Piano 3       | 8: Piano 3                                            |
| 3                           | Honky-tonk    | 8: Honky-tonk                                         |
| 4                           | E.Piano 1     | 8: Detuned EP 1<br>16: E.Piano 1v<br>24: 60's E.Piano |
| 5                           | E.Piano 2     | 8: Detuned EP 2<br>16: E.Piano 2v                     |
| 6                           | Harpsichord   | 8: Coupled Hps.<br>16: Harpsichord<br>24: Harpsi.o    |
| 7                           | Clav.         | —                                                     |
| <b>Chromatic Percussion</b> |               |                                                       |
| 8                           | Celesta       | —                                                     |
| 9                           | Glockenspiel  | —                                                     |
| 10                          | Music Box     | —                                                     |
| 11                          | Vibraphone    | 8: Vibraphone                                         |
| 12                          | Marimba       | 8: Marimba                                            |
| 13                          | Xylophone     | —                                                     |
| 14                          | Tubular-bell  | 8: Church Bell<br>9: Carillon                         |
| 15                          | Santur        | —                                                     |
| <b>Organ</b>                |               |                                                       |
| 16                          | Organ 1       | 8: Detuned Or.1<br>16: 60's Organ 1<br>32: Organ 4    |
| 17                          | Organ 2       | 8: Detuned Or.2<br>32: Organ 5                        |
| 18                          | Organ 3       | —                                                     |
| 19                          | Church Org.1  | 8: Church Org.2<br>16: Church Org.3                   |
| 20                          | Reed Organ    | —                                                     |
| 21                          | Accordion Fr  | 8: Accordion It                                       |
| 22                          | Harmonica     | —                                                     |
| 23                          | Bandoneon     | —                                                     |
| <b>Guitar</b>               |               |                                                       |
| 24                          | Nylon-str.Gt  | 8: Ukulele<br>16: Nylon Gt.o<br>32: Nylon Gt.2        |
| 25                          | Steel-str.Gt  | 8: 12-str.Gt<br>16: Mandolin                          |
| 26                          | Jazz Gt.      | 8: Hawaiian Gt.                                       |
| 27                          | Clean Gt.     | 8: Chorus Gt.                                         |
| 28                          | Muted Gt.     | 8: Funk Gt.<br>16: Funk Gt.2                          |
| 29                          | Overdrive Gt  | —                                                     |
| 30                          | DistortionGt  | 8: Feedback Gt.                                       |
| 31                          | Gt.Harmonics  | 8: Gt. Feedback                                       |
| <b>Bass</b>                 |               |                                                       |
| 32                          | Acoustic Bs.  | —                                                     |
| 33                          | Fingered Bs.  | —                                                     |
| 34                          | Picked Bs.    | —                                                     |

| #               | Name (bank 0) | Variants (bank MSB: name)           |
|-----------------|---------------|-------------------------------------|
| 35              | Fretless Bs.  | —                                   |
| 36              | Slap Bass 1   | —                                   |
| 37              | Slap Bass 2   | —                                   |
| 38              | Synth Bass 1  | 1: SynthBass101<br>8: Synth Bass 3  |
| 39              | Synth Bass 2  | 8: Synth Bass 4<br>16: Rubber Bass  |
| <b>Strings</b>  |               |                                     |
| 40              | Violin        | 8: Slow Violin                      |
| 41              | Viola         | —                                   |
| 42              | Cello         | —                                   |
| 43              | Contrabass    | —                                   |
| 44              | Tremolo Str   | —                                   |
| 45              | PizzicatoStr  | —                                   |
| 46              | Harp          | —                                   |
| 47              | Timpani       | —                                   |
| <b>Ensemble</b> |               |                                     |
| 48              | Strings       | 8: Orchestra                        |
| 49              | Slow Strings  | —                                   |
| 50              | Syn.Strings1  | 8: Syn.Strings3                     |
| 51              | Syn.Strings2  | —                                   |
| 52              | Choir Aahs    | 32: Choir Aahs 2                    |
| 53              | Voice Oohs    | —                                   |
| 54              | SynVox        | —                                   |
| 55              | OrchestraHit  | —                                   |
| <b>Brass</b>    |               |                                     |
| 56              | Trumpet       | —                                   |
| 57              | Trombone      | 1: Trombone 2                       |
| 58              | Tuba          | —                                   |
| 59              | MutedTrumpet  | —                                   |
| 60              | French Horns  | 1: Fr.Horn 2                        |
| 61              | Brass 1       | 8: Brass 2                          |
| 62              | Synth Brass1  | 8: Synth Brass3<br>16: AnalogBrass1 |
| 63              | Synth Brass2  | 8: Synth Brass4<br>16: AnalogBrass2 |
| <b>Reed</b>     |               |                                     |
| 64              | Soprano Sax   | —                                   |
| 65              | Alto Sax      | —                                   |
| 66              | Tenor Sax     | —                                   |
| 67              | Baritone Sax  | —                                   |
| 68              | Oboe          | —                                   |
| 69              | English Horn  | —                                   |
| 70              | Bassoon       | —                                   |
| 71              | Clarinet      | —                                   |
| <b>Pipe</b>     |               |                                     |
| 72              | Piccolo       | —                                   |
| 73              | Flute         | —                                   |
| 74              | Recorder      | —                                   |
| 75              | Pan Flute     | —                                   |
| 76              | Bottle Blow   | —                                   |

| #                    | Name (bank 0) | Variants (bank MSB: name)   |
|----------------------|---------------|-----------------------------|
| 77                   | Shakuhachi    | —                           |
| 78                   | Whistle       | —                           |
| 79                   | Ocarina       | —                           |
| <b>Synth Lead</b>    |               |                             |
| 80                   | Square Wave   | 1: Square<br>8: Sine Wave   |
| 81                   | Saw Wave      | 1: Saw<br>8: Doctor Solo    |
| 82                   | Syn.Calliope  | —                           |
| 83                   | Chiffer Lead  | —                           |
| 84                   | Charang       | —                           |
| 85                   | Solo Vox      | —                           |
| 86                   | 5th Saw Wave  | —                           |
| 87                   | Bass & Lead   | —                           |
| <b>Synth Pad</b>     |               |                             |
| 88                   | Fantasia      | —                           |
| 89                   | Warm Pad      | —                           |
| 90                   | Polysynth     | —                           |
| 91                   | Space Voice   | —                           |
| 92                   | Bowed Glass   | —                           |
| 93                   | Metal Pad     | —                           |
| 94                   | Halo Pad      | —                           |
| 95                   | Sweep Pad     | —                           |
| <b>Synth Effects</b> |               |                             |
| 96                   | Ice Rain      | —                           |
| 97                   | Soundtrack    | —                           |
| 98                   | Crystal       | 1: Syn Mallet               |
| 99                   | Atmosphere    | —                           |
| 100                  | Brightness    | —                           |
| 101                  | Goblin        | —                           |
| 102                  | Echo Drops    | 1: Echo Bell<br>2: Echo Pan |
| 103                  | Star Theme    | —                           |
| <b>Ethnic</b>        |               |                             |
| 104                  | Sitar         | 1: Sitar 2                  |
| 105                  | Banjo         | —                           |
| 106                  | Shamisen      | —                           |
| 107                  | Koto          | 8: Taisho Koto              |
| 108                  | Kalimba       | —                           |
| 109                  | Bagpipe       | —                           |
| 110                  | Fiddle        | —                           |
| 111                  | Shanai        | —                           |
| <b>Percussive</b>    |               |                             |
| 112                  | Tinkle Bell   | —                           |
| 113                  | Agogo         | —                           |
| 114                  | Steel Drums   | —                           |
| 115                  | Woodblock     | 8: Castanets                |
| 116                  | Taiko         | 8: Concert BD               |
| 117                  | Melo. Tom 1   | 8: Melo. Tom 2              |
| 118                  | Synth Drum    | 8: 808 Tom<br>9: Elec Perc. |

| #                    | Name (bank 0) | Variants (bank MSB: name)                                                                                                           |
|----------------------|---------------|-------------------------------------------------------------------------------------------------------------------------------------|
| 119                  | Reverse Cym.  | —                                                                                                                                   |
| <b>Sound Effects</b> |               |                                                                                                                                     |
| 120                  | Gt.FretNoise  | 1: Gt.Cut Noise<br>2: String Slap                                                                                                   |
| 121                  | Breath Noise  | 1: Fl.Key Click                                                                                                                     |
| 122                  | Seashore      | 1: Rain<br>2: Thunder<br>3: Wind<br>4: Stream<br>5: Bubble                                                                          |
| 123                  | Bird          | 1: Dog<br>2: Horse-Gallop<br>3: Bird 2                                                                                              |
| 124                  | Telephone 1   | 1: Telephone 2<br>2: DoorCreaking<br>3: Door<br>4: Scratch<br>5: Wind Chimes                                                        |
| 125                  | Helicopter    | 1: Car-Engine<br>2: Car-Stop<br>3: Car-Pass<br>4: Car-Crash<br>5: Siren<br>6: Train<br>7: Jetplane<br>8: Starship<br>9: Burst Noise |
| 126                  | Applause      | 1: Laughing<br>2: Screaming<br>3: Punch<br>4: Heart Beat<br>5: Footsteps                                                            |
| 127                  | Gun Shot      | 1: Machine Gun<br>2: Lasergun<br>3: Explosion                                                                                       |

## 5.2. Drums

### NOTE

For a Part in rhythm mode, note numbers select individual drum/percussion sounds instead of a pitched instrument — but which sound each note number produces differs per drum kit.

Available kits:

| Program | Kit        |
|---------|------------|
| 0       | Standard   |
| 8       | Room       |
| 16      | Power      |
| 24      | Electronic |
| 25      | TR-808     |
| 32      | Jazz       |
| 40      | Brush      |
| 48      | Orchestra  |
| 56      | SFX        |

The Standard kit covers note numbers **27–87** with no gaps. The GM percussion assignments for 35–81 are:

| Note | Sound              | Note | Sound          |
|------|--------------------|------|----------------|
| 35   | Acoustic Bass Drum | 59   | Ride Cymbal 2  |
| 36   | Bass Drum 1        | 60   | Hi Bongo       |
| 37   | Side Stick         | 61   | Low Bongo      |
| 38   | Acoustic Snare     | 62   | Mute Hi Conga  |
| 39   | Hand Clap          | 63   | Open Hi Conga  |
| 40   | Electric Snare     | 64   | Low Conga      |
| 41   | Low Floor Tom      | 65   | High Timbale   |
| 42   | Closed Hi-Hat      | 66   | Low Timbale    |
| 43   | High Floor Tom     | 67   | High Agogo     |
| 44   | Pedal Hi-Hat       | 68   | Low Agogo      |
| 45   | Low Tom            | 69   | Cabasa         |
| 46   | Open Hi-Hat        | 70   | Maracas        |
| 47   | Low-Mid Tom        | 71   | Short Whistle  |
| 48   | Hi-Mid Tom         | 72   | Long Whistle   |
| 49   | Crash Cymbal 1     | 73   | Short Guiro    |
| 50   | High Tom           | 74   | Long Guiro     |
| 51   | Ride Cymbal 1      | 75   | Claves         |
| 52   | Chinese Cymbal     | 76   | Hi Wood Block  |
| 53   | Ride Bell          | 77   | Low Wood Block |
| 54   | Tambourine         | 78   | Mute Cuica     |
| 55   | Splash Cymbal      | 79   | Open Cuica     |
| 56   | Cowbell            | 80   | Mute Triangle  |
| 57   | Crash Cymbal 2     | 81   | Open Triangle  |
| 58   | Vibraslap          |      |                |

Note numbers 27–34 and 82–87 also sound; they fall outside the GM percussion map and what they play is a property of the loaded DLS file. Note numbers outside 27–87 are **silent** — no region covers them, and the note is dropped rather than substituted with a nearby sound.